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On Compensation Function Observer-Based Active Disturbance Rejection Control for MIMO Uncertain Systems and Its Application to 2-DOF Helicopter System

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***Abstract*—This article investigates the control problem for multiple-input multiple-output systems with nonlinear uncertainties. By introducing a prediction-correction mechanism, a new compensation function observer (CFO) is constructed to enhance the estimation accuracy of the total disturbance. Subsequently, the CFO-based active disturbance rejection control (CADRC) is designed to mitigate the total disturbances and attain the desired control performance. Based on an inductive decoupling-based Lyapunov method, this article rigorously establishes the closed-loop stability of the CADRC under a broad class of nonlinear state-dependent uncertainties. Moreover, a comparative frequency-domain analysis quantitatively reveals the superior disturbance estimation and rejection capabilities of the CADRC. Finally, the proposed CADRC is implemented on a Quanser two-degree-of-freedom helicopter system subject to additional digital uncertainties, and the experimental results demonstrate its clear superiority over the conventional ADRC.**

***Index Terms*—Active disturbance rejection control (ADRC), compensation function observer (CFO), multiple-input multiple-output system (MIMO), uncertain system.**

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I. Introduction

SINCE uncertainties are ubiquitous in physical systems, it is one of the fundamental issues in the control field to tackle these uncertainties [1]. To preserve the normal system operation in the presence of various uncertainties, abundant control strategies have been proposed and developed, such as proportional-integral-derivative (PID) control [2], active disturbance rejection control (ADRC) [3], [4], disturbance observer-based control [5], [6], generalized proportional integral observer-based control [7], [8], embedded model control [9], [10], model predictive control [11], finite-time control [12], [13], predefined-time control [14], etc. Among these methods, ADRC has attracted considerable attention across academia and industry due to its profound theoretical foundation and exceptional practical performance.

The core principle of ADRC lies in constructing an extended state observer (ESO) to actively estimate the total disturbance [15], which is subsequently compensated for within the closed-loop system. By formulating disturbances in a generalized framework and leveraging the strong estimation capability of the ESO, ADRC can effectively cope with a wide range of uncertainties and achieve satisfactory closed-loop performance. To date, ADRC has found successful application in a variety of industrial systems, such as PMSM system [16], power system [17], robotic system [18], motion system [19], magnetic levitation system [20], etc. Owing to the pivotal role of ESO, substantial research has focused on enhancing its estimation accuracy and improving its overall performance [21], [22], [23], [24], [25], [26], [27]. Recently, Qi et al. [28] pointed out the limitations of the traditional ESO, such as nonderivative structure and insufficient usage of information. To tackle these limitations, they first proposed a novel third order compensation function observer (CFO) for a class of second order uncertain systems. In the design framework of CFO, the derivative form, measurable innovation, and an explicit compensator are adopted to enhance the estimation performance, which has been validated through both theoretical analysis and experimental studies.

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In the development of CFO, several notable advances have been made. The reference [29] developed a general design of the CFO for a class of nth order single-input single-output (SISO) uncertain systems, and demonstrated its advantages in the disturbance estimation accuracy and estimable bandwidth. Wang et al. [30] investigated the application of CFO in PMSM system, and highlighted the merits of the CFO-based ADRC in disturbance rejection capability. The reference [31] explored the implementation of CFO in the quadrotor UAV system, and effectively addressed the dependence on modeling accuracy in traditional backstepping control. Nevertheless, the general design of the CFO-based ADRC (CADRC) for multiple-input multiple-output (MIMO) uncertain system remains unexplored. Furthermore, the inherent complexity of parameter design in traditional CFO structures requires further investigation and simplification.

As for the theoretical foundations of the CFO, the existing studies remain incomplete and warrant further development. Under the assumption that the total disturbance is a $(n+1)$th order infinitesimal function, the reference [29] rigorously proved the convergence of the $(n+1)$th order CFO for a class of nth order SISO uncertain systems. However, in practical scenarios, the total disturbance often depends on the system states [15], which makes it challenging to precisely analyze the estimation performance of CFO by existing methods [28], [29]. Moreover, as reported in [29], the closed-loop stability of the CADRC remains unresolved, which calls for further investigation to ensure its reliable application. The main difficulties in the theoretical analysis of CADRC under nonlinear uncertainties can be summarized as the following two points.

(D1) Since the uncertainties are nonlinearly dependent on the system states, it is sometimes unreasonable to directly assume that the uncertainties and their derivatives are bounded. Moreover, the strong cross-channel coupling exists in the MIMO systems, where the uncertainty in any single channel is affected by all states of the control system. Therefore, it is challenging to characterize and prove the scope of uncertainties that the CADRC can effectively suppress.

(D2) Due to the complex coupling effect between the control system and the observer system, it is insufficient to conduct a standalone convergence analysis of CFO, such as [28], [29]. How to comprehensively analyze the closed-loop transient performance of CADRC under nonlinear state-dependent uncertainties remains a significant theoretical hurdle.

To address the aforementioned issues, this article is devoted to the design and theoretical analysis of CADRC for a class of MIMO systems with nonlinear uncertainties. Furthermore, the proposed CADRC is applied to a Quanser two-degree-of-freedom (2-DOF) helicopter system with additional digital uncertainties. The key contribution is outlined as follows.

1) A new CADRC is proposed for a class of MIMO uncertain systems. In the proposed design, the prediction-correction mechanism is introduced to enhance the estimation accuracy of the total disturbances. Moreover, by investigating the estimation errors of the CFO, a single-parameter-tuning strategy is devised to simplify the parameter design.
2) The closed-loop stability of the CADRC-based MIMO system is rigorously investigated in the presence of a broad class of nonlinear state-dependent uncertainties. Moreover, it is shown that the tracking and estimation errors can converge to the desired bound by increasing the bandwidth of CFO.
3) The transfer functions from the total disturbance to both its estimation error and the controlled output are explicitly derived. Then, the frequency response analysis of these transfer functions quantitatively reveals the superiority of CADRC in both disturbance estimation and rejection capabilities.

The remainder of this article is structured as follows. Section II provides the problem formulation. Section III presents the design of the ADRC with a single-parameter-tuning CFO. The performance analyses in both time domain and frequency domain are provided in Section IV. The experiments are discussed in Section V. Section VI concludes the article.

II. Problem Formulation

Consider the following multiple-input multiple-output (MIMO) nonlinear uncertain system

$$\begin{cases} \dot{x}_{i1}(t) = x_{i2}(t), \\ \vdots \\ \dot{x}_{i(n-1)}(t) = x_{in}(t), \\ \dot{x}_{in}(t) = \delta_i(x(t),t) + d_i(t) + \sum\limits_{j=1}^{m} b_{ij}(t)u_j(t), \\ y_i(t) = Cx_i(t) = x_{i1}(t), \quad t \geqslant t_0, \quad i = 1,2,\ldots,m, \end{cases} \tag{1}$$

where $x(t) = [x_1^\top(t), \ldots, x_m^\top(t)]^\top \in \mathbb{R}^{nm}$, $x_i(t) = [x_{i1}(t), \ldots, x_{in}(t)]^\top \in \mathbb{R}^n$ represent the state vector, $u_j(t) \in \mathbb{R}$ represents the control input, $y_i(t) \in \mathbb{R}$ represents the controlled output, $d_i(t) \in \mathbb{R}$ represents the time-varying external disturbances, and $\delta_i(x(t),t) \in \mathbb{R}$ represents the coupled internal uncertainties. Additionally, the known control input gain matrix $B_u(t) = [b_{ij}(t)]_{m\times m} \in \mathbb{R}^{m\times m}$ is invertible, and the output matrix is defined as $C = [1, 0, \ldots, 0] \in \mathbb{R}^{1\times n}$.

Remark 1: It is worth emphasizing that each internal uncertainty $\delta_i(x(t),t)$ is a nonlinear function of the entire state vector $x(t)$, resulting in strong cross-channel coupling.

The control target is to design the control input $u(t) \triangleq [u_1(t), \ldots, u_m(t)]^\top \in \mathbb{R}^m$ such that the output $y(t) \triangleq [y_1(t), \ldots, y_m(t)]^\top \in \mathbb{R}^m$ tracks a given reference signal $r(t) \triangleq [r_1(t), \ldots, r_m(t)]^\top \in \mathbb{R}^m$. Additionally, the following assumptions are presented regarding the unknown nonlinear functions $\delta_i(\cdot)$, $d_i(\cdot)$, and the reference signal $r_i(t)$, $i \in \underline{m}$.

Assumption 1: The function $\delta_i(\cdot) \in \mathcal{C}^{n+1}(\mathbb{R}^{nm} \times \mathbb{R}, \mathbb{R})$. Furthermore, there exists a function $\varphi(\cdot) \in \mathcal{C}(\mathbb{R}^{nm}, [0, +\infty))$ such that for any integers $i_x \geqslant 0$, $i_t \geqslant 0$, and $i_x + i_t \leqslant n+1$

$$\sup_{t\geqslant t_0} \left| \frac{\partial^{i_x+i_t}\delta_i}{\partial x^{i_x}\partial t^{i_t}} \right| \leqslant \varphi(x), \quad \forall x \in \mathbb{R}^{nm}. \tag{2}$$

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Assumption 2: [5], [24] The function $d_i(\cdot) \in \mathcal{C}^{n+1}(\mathbb{R},\mathbb{R})$. Furthermore, there exists a positive constant N_d such that

$$\sup_{t\geqslant t_0}\left|d_i^{(n+1)}(t)\right| \leqslant N_d. \tag{3}$$

Remark 2: Under Assumption 1, the nonlinear function $\delta_i(x,t)$ and its partial derivatives $(\partial^{i_x+i_t}\delta_i/\partial x^{i_x}\partial t^{i_t})$ are uniformly bounded by a continuous function $\varphi(x)$, which covers a broad class of uncertainties. For Assumption 2, the boundedness of the $(n+1)$th order derivative of the nonlinear function $d_i(t)$ is required. Additionally, if $d_i(t)$ is a discontinuous function with the first class of discontinuous points, the stability can still be established in a piecewise manner, as shown in [32].

Assumption 3: [5], [15] The function $r_i(\cdot) \in \mathcal{C}^{2n+1}(\mathbb{R},\mathbb{R})$. Furthermore, there exists a positive constant N_r such that

$$\sup_{t\geqslant t_0}\left\{|r_i(t)|, \left|r_i^{(1)}(t)\right|, \ldots, \left|r_i^{(2n+1)}(t)\right|\right\} \leqslant N_r. \tag{4}$$

From Assumption 3, the reference signal $r(t)$ and its derivatives up to $(2n+1)$ order are assumed to be bounded, which are typical in practical engineering scenarios.

III. Design of CADRC

In this section, the CFO-based ADRC is designed for the MIMO system (1). First, a single-parameter-tuning CFO is designed to estimate the total disturbance for each channel individually. Then, based on these estimations, the overall feedback controller is constructed. Let $g_i(x(t),t) \triangleq \delta_i(x(t),t) + d_i(t)$ denote the total disturbance, and let $u_i^*(t) \triangleq \sum_{j=1}^m b_{ij}(t)u_j(t)$ be the virtual input, for $i = 1,2,\ldots,m$.

From the system (1), the state vector $x_i(t)$ is governed by the following equation:

$$\dot{x}_i(t) = Ax_i(t) + B\left(g_i(x(t),t) + u_i^*(t)\right) \tag{5}$$

where the matrices A and B are defined as $A = \begin{bmatrix} \mathbf{0}_{(n-1)\times 1} & I_{(n-1)\times(n-1)} \\ \mathbf{0}_{1\times 1} & \mathbf{0}_{1\times(n-1)} \end{bmatrix} \in \mathbb{R}^{n\times n}$, $B = \begin{bmatrix} \mathbf{0}_{1\times(n-1)} & 1 \end{bmatrix}^T \in \mathbb{R}^n$.

Then, a single-parameter-tuning CFO is designed to estimate the total disturbance $g_i(x(t),t)$ through the following steps. In addition, define the extended state vector as $x_{ie}(t) = [x_{i1},\ldots,x_{i(n+1)}]^\top \triangleq \left[x_i^\top(t)\ g_i(x(t),t)\right]^\top \in \mathbb{R}^{n+1}$.

Step 1 (Prediction): To begin with, we provide a preliminary estimation for the total disturbance $x_{i(n+1)}(t)$

$$\begin{cases} \dot{\hat{x}}_i(t) = A\hat{x}_i(t) + B\left(z_i(t) + L_1^\top(x_i(t) - \hat{x}_i(t)) + u_i^*(t)\right), \\ \dot{z}_i(t) = L_2^\top\left(x_i(t) - \hat{x}_i(t)\right) \end{cases} \tag{6}$$

where $\hat{x}_i(t) = [\hat{x}_{i1}(t),\ldots,\hat{x}_{in}(t)]^\top \in \mathbb{R}^n$ is the estimation of $x_i(t)$, $z_i(t) \in \mathbb{R}$ is preliminary estimation of $x_{i(n+1)}(t)$, and $L_1 = [l_{11},\ldots,l_{1n}]^\top \in \mathbb{R}^n$, $L_2 = [l_{21},\ldots,l_{2n}]^\top \in \mathbb{R}^n$ are the parameter vectors to be designed.

Step 2 (Correction): Then, to enhance the estimation accuracy of the total disturbance, the preliminary estimation $z_i(t)$ is refined by estimation errors $(x_i(t) - \hat{x}_i(t))$. The actual estimation of the total disturbance is obtained by the following equation:

$$\hat{x}_{i(n+1)}(t) = z_i(t) + L_1^\top\left(x_i(t) - \hat{x}_i(t)\right) \tag{7}$$

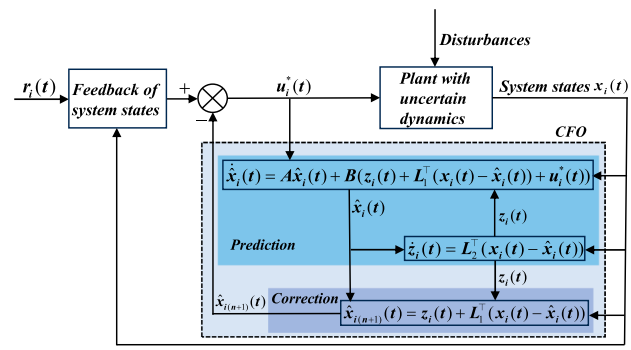

Fig. 1. Block diagram of the proposed CADRC scheme for each channel.

where $\hat{x}_{i(n+1)}(t) \in \mathbb{R}$ represents the estimation of $x_{i(n+1)}(t)$.

Step 3 (Parameter Design): Next, to facilitate parameter design, the estimation errors of CFO (6) and (7) are investigated. By taking the derivative of $\hat{x}_{i(n+1)}(t)$, the CFO (6) and (7) can be equivalently rewritten as follows:

$$\dot{\hat{x}}_{ie}(t) = A_e\hat{x}_{ie}(t) + B_e u_i^*(t) + D_e L_e^\top\left(x_{ie}(t) - \hat{x}_{ie}(t)\right) \tag{8}$$

where $\hat{x}_{ie}(t) = [\hat{x}_i^\top(t), \hat{x}_{i(n+1)}(t)]^\top \in \mathbb{R}^{n+1}$, the parameter vector $L_e = [l_{21}, l_{22}+l_{11},\ldots,l_{2n}+l_{1(n-1)}, l_{1n}]^\top \in \mathbb{R}^{n+1}$, and the matrices A_e, B_e, D_e are defined by $A_e = \begin{bmatrix} \mathbf{0}_{n\times 1} & I_{n\times n} \\ \mathbf{0}_{1\times 1} & \mathbf{0}_{1\times n} \end{bmatrix} \in \mathbb{R}^{(n+1)\times(n+1)}$, $B_e = \begin{bmatrix} \mathbf{0}_{1\times(n-1)} & 1 & 0 \end{bmatrix}^\top \in \mathbb{R}^{n+1}$, $D_e = \begin{bmatrix} \mathbf{0}_{1\times n} & 1 \end{bmatrix}^\top \in \mathbb{R}^{n+1}$.

Combining (5) and (8), we derive that the estimation errors of CFO $e_i(t) \triangleq x_{ie}(t) - \hat{x}_{ie}(t)$ satisfy the following equation:

$$\dot{e}_i(t) = \left(A_e - D_e L_e^\top\right)e_i(t) + D_e\dot{g}_i(x(t),t). \tag{9}$$

To ensure the stability of (9), the parameter vector L_e is designed to render the matrix $\left(A_e - D_e L_e^\top\right)$ Hurwitz. Specifically, by using the bandwidth theory in [21], we design

$$\begin{aligned} \left|sI_{n+1} - A_e + D_e L_e^\top\right| &= s^{n+1} + l_{1n}s^n \\ &+ \cdots + (l_{11}+l_{22})s + l_{21} = (s+\mu)^{n+1} \end{aligned} \tag{10}$$

where $\mu > 0$ is viewed as the bandwidth of CFO. Then, by setting $L_1 = [(1/2)l_n,\ldots,(1/2)l_2,l_1]^\top$ and $L_2 = [l_{n+1},(1/2)l_n,\ldots,(1/2)l_2]^\top$ with $l_j = C_{n+1}^j\mu^j$, $j \in \underline{n+1}$, the parameters can be uniquely determined with respect to the given bandwidth μ, leaving only one parameter to be tuned. The general scheme of CFO is presented in Fig. 1.

Remark 3: According to the traditional CFO in [28], [29], $z_i(t)$ serves as the compensation function. Unlike the previous design, $z_i(t)$ is defined as the integral of $L_2^\top(x_i(t) - \hat{x}_i(t))$, rather than the integral of $\lambda L_1^\top(x_i(t) - \hat{x}_i(t))$. Moreover, under the proposed design (10), the parameter design of the CFO is simplified to tuning a single bandwidth μ, which promotes its industrial application.

Finally, based on (6) and (7), the CFO-based feedback controller $u(t)$ is designed as follows:

$$\begin{cases} u_i^*(t) = r_i^{(n)}(t) - K^\top\left(x_i(t) - \bar{r}_i(t)\right) - \hat{x}_{i(n+1)}(t), \ i \in \underline{m} \\ u(t) = B_u^{-1}(t)u^*(t) \triangleq B_u^{-1}(t)[u_1^*(t),\ldots,u_m^*(t)]^\top \end{cases} \tag{11}$$

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where $\bar{r}_i(t) = [r_i(t), r_i^{(1)}(t), \ldots, r_i^{(n-1)}(t)]^\top \in \mathbb{R}^n$, and the control gain vector $K = [k_1, \ldots, k_n]^\top \in \mathbb{R}^n$ is chosen to make the matrix $A_K \triangleq A - BK^\top$ Hurwitz.

IV. Performance Analysis

A. Time-Domain Performance Analysis

In this section, the closed-loop stability of the CADRC is analyzed. The analysis unfolds in the following two steps: 1) establishing an equivalent realization of the CADRC; and 2) proving the stability of the CADRC-based closed-loop system.

First, to establish a benchmark for the disturbance rejection performance, we design the following ideal trajectory

$$\begin{cases} \dot{x}_i^*(t) = Ax_i^*(t) - BK^\top(x_i^*(t) - \bar{r}_i(t)) + Br_i^{(n)}(t), \\ y_i^*(t) = C^\top x_i^*(t), \quad t \geqslant t_0, \quad x_i^*(t_0) = x_i(t_0) \end{cases} \tag{12}$$

where $x_i^*(t) = [x_{i1}^*(t), \ldots, x_{in}^*(t)]^\top \in \mathbb{R}^n$ represents the ideal state vector, and $y_i^*(t) \in \mathbb{R}$ represents the ideal output, which exponentially converges to the reference signal $r_i(t)$ at a rate governed by the parameter vector K.

Then, an equivalent realization of the CADRC (6), (7), and (11) is introduced to facilitate the theoretical analysis.

Lemma 1: For the system (5), an equivalent realization of the CADRC (6), (7), and (11) is established as

$$\begin{cases} \dot{\hat{G}}_i(t) = A_e\hat{G}_i(t) + \tilde{L}_eC_e^\top\left(G_i(t) - \hat{G}_i(t)\right), \\ \tilde{u}_i^*(t) = r_i^{(n)}(t) - K^\top(x_i(t) - \bar{r}_i(t)) - \hat{g}_{i1}(t), \ i \in \underline{m} \\ \tilde{u}(t) = B_u^{-1}(t)\tilde{u}^*(t) \triangleq B_u^{-1}(t)\left[\tilde{u}_1^*(t), \ldots, \tilde{u}_m^*(t)\right]^\top \end{cases} \tag{13}$$

where $C_e = \left[1 \ \mathbf{0}_{1\times n}\right]^\top \in \mathbb{R}^{n+1}$, the parameter vector $\tilde{L}_e = [l_1, \ldots, l_{n+1}]^\top \in \mathbb{R}^{n+1}$, and $\hat{G}_i(t) = [\hat{g}_{i1}(t), \ldots, \hat{g}_{i(n+1)}(t)]^\top \in \mathbb{R}^{n+1}$ is the estimation of $G_i(t) = [g_i(x(t), t), g_i^{(1)}(x(t), t), \ldots, g_i^{(n)}(x(t), t)]^\top \in \mathbb{R}^{n+1}$.

The proof of Lemma 1 is presented in Appendix A. Based on the realization in (13), the following theorem demonstrates the closed-loop performance of the CADRC-based MIMO system under nonlinear state-dependent coupling uncertainties.

Theorem 1: Consider the CADRC-based closed-loop system (1), (6), (7), and (11). Suppose that Assumptions 1–3 are satisfied, and the matrix A_K is Hurwitz. Then, there exist positive constants $\tilde{\mu}$ and $\rho_j (1 \leqslant j \leqslant 3)$ dependent on $(\varphi(\cdot), \ N_r, \ N_d)$, such that if $\mu \in [\tilde{\mu}, +\infty)$, then

$$\sup_{t\geqslant t_0} \|x_i(t) - x_i^*(t)\| \leqslant \rho_1 \frac{ln\mu}{\mu} \tag{14}$$

$$\sup_{t\geqslant \tilde{t}} \left\|G_i(t) - \hat{G}_i(t)\right\| \leqslant \rho_2\left(e^{-\rho_3\mu(t-\tilde{t})} + \frac{1}{\mu}\right) \tag{15}$$

for any $i \in \underline{m}$, where the constants $\rho_j (1 \leqslant j \leqslant 3)$ are μ-independent, $\tilde{t} = t_0 + 2n\lambda_{\phi 2}(ln\mu/\mu)$, and $\lambda_{\phi 2}$ is given in (26) in Appendix B.

The proof of Theorem 1 is presented in Appendix B. As shown in (14) and (15), Theorem 1 establishes the explicit and tunable upper bounds for the tracking and estimation errors in the CADRC-based closed-loop system. From (14), by selecting a sufficiently large observer bandwidth μ, the tracking error between the actual and ideal trajectories can be arbitrarily small in the whole time domain. According (15), the estimation errors can exponentially converge to the desired bound by increasing the observer bandwidth μ.

Remark 4: As demonstrated in Theorem 1, the desired closed-loop performance in the presence of nonlinear state-dependent uncertainties can be attained by increasing μ. Moreover, the relationship between the closed-loop performance and the controller parameters is explicitly depicted. The controller gain K is selected to determine the ideal tracking performance, and the observer bandwidth μ is increased to enhance the disturbance rejection capability, which provides guidance for practical engineering applications.

B. Frequency-Domain Performance Analysis

In this section, a frequency-domain performance analysis of the proposed CADRC is presented.

First, we define $Y_i(s)$, $R_i(s)$, $X_{i(n+1)}(s)$, and $\hat{X}_{i(n+1)}(s)$ as the Laplace transforms of the system output $y_i(t)$, the reference signal $r_i(t)$, the total disturbance $x_{i(n+1)}(t)$, and its corresponding estimation $\hat{x}_{i(n+1)}(t)$, respectively. Then, the transfer functions from the total disturbance to both its estimation error and the controlled output are explicitly derived, as presented in Proposition 1.

Proposition 1: Consider the system (1) under two control methods: 1) the CADRC (6)–(11); and 2) the LADRC [33]. Suppose zero initial condition, then

$$\begin{cases} X_{i(n+1)}(s) - \hat{X}_{i(n+1),p}(s) = L_{1,p}(s)X_{i(n+1)}(s), \\ Y_{i,p}(s) = L_{2,p}(s)X_{i(n+1)}(s) + R_i(s), \quad p = a, b, \end{cases} \tag{16}$$

where the index p denotes the control method, and the transfer functions $L_{1,p}(s)$ and $L_{2,p}(s)$ are defined as

$$\begin{cases} L_{1,a}(s) = \frac{s^{n+1}}{s^{n+1}+\sum_{p=1}^{n+1} l_p s^{n+1-p}}, \\ L_{2,a}(s) = \frac{s^{n+1}}{\left(s^{n+1}+\sum_{p=1}^{n+1} l_p s^{n+1-p}\right)\left(s^{n}+\sum_{p=0}^{n-1} k_{n-p} s^{n-1-p}\right)}, \\ L_{1,b}(s) = \frac{s^{n+1}+\sum_{p=1}^{n} l_p s^{n+1-p}}{s^{n+1}+\sum_{p=1}^{n+1} l_p s^{n+1-p}}, \\ L_{2,b}(s) = \frac{s^{n+1}+\sum_{p=1}^{n}\left(l_p+\sum_{q=1}^{p-1} k_{n+1-q}l_{p-q}+k_{n+1-p}\right)s^{n+1-p}}{\left(s^{n+1}+\sum_{p=1}^{n+1} l_p s^{n+1-p}\right)\left(s^{n}+\sum_{p=0}^{n-1} k_{n-p} s^{n-1-p}\right)}. \end{cases}$$

The proof of Proposition 1 is provided in Appendix C. By analyzing the frequency response of $L_{1,a}(s)$ and $L_{2,a}(s)$ in (16), the disturbance estimation and rejection capabilities of CADRC are assessed. Furthermore, a comparative result between the proposed CADRC and the traditional LADRC is established in Proposition 2.

Proposition 2: Consider the system (1) under two control methods: (a) the CADRC (6)–(11), and (b) the LADRC [33]. Suppose that the initial values are zero, and A_K is Hurwitz. Then, for any bandwidth $\mu > 0$, the following relationships hold

$$\lim_{w\to 0} |L_{1,a}(jw)| = 0, \quad \lim_{w\to 0} |L_{2,a}(jw)| = 0 \tag{17}$$

$$\lim_{w\to 0} \left|\frac{L_{1,a}(jw)}{L_{1,b}(jw)}\right| = 0, \quad \lim_{w\to 0} \left|\frac{L_{2,a}(jw)}{L_{2,b}(jw)}\right| = 0 \tag{18}$$

where j represents the imaginary unit.

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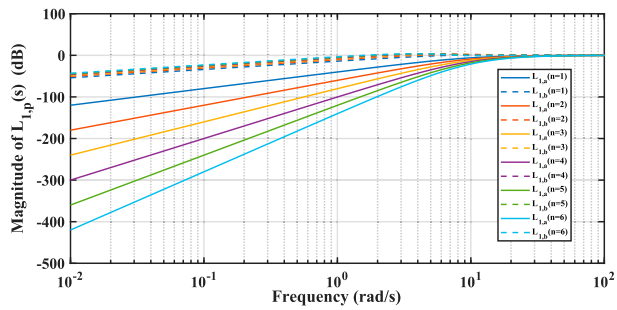

Fig. 2. Magnitude response curves of $L_{1,a}(s)$ and $L_{1,b}(s)$ under the parameter configuration $(\mu = 10)$.

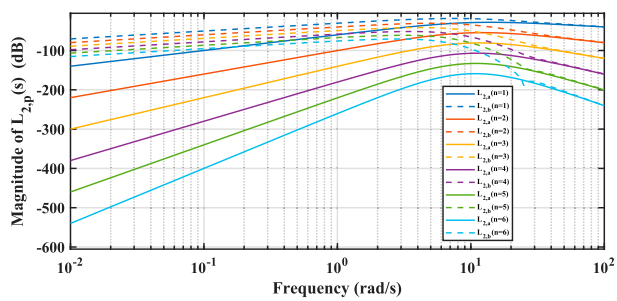

Fig. 3. Magnitude response curves of $L_{2,a}(s)$ and $L_{2,b}(s)$ under the parameter configuration $(\mu = 10, \omega_c = 10)$.

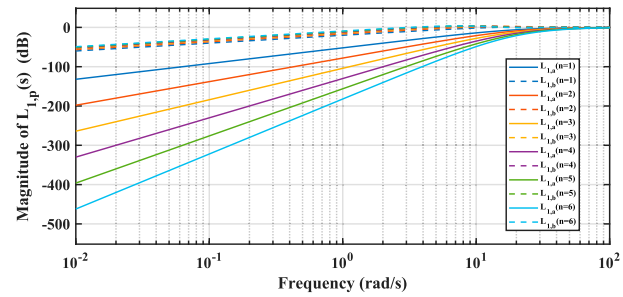

Fig. 4. Magnitude response curves of $L_{2,a}(s)$ and $L_{2,b}(s)$ under the parameter configuration $(\mu = 20)$.

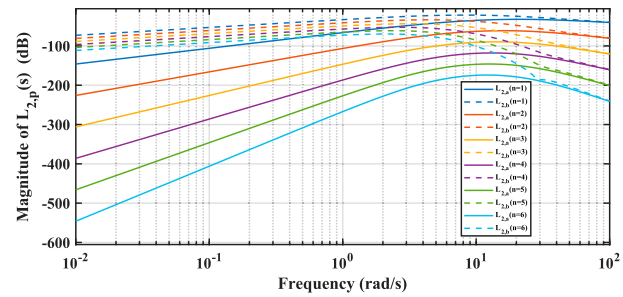

Fig. 5. Magnitude response curves of $L_{2,a}(s)$ and $L_{2,b}(s)$ under the parameter configuration $(\mu = 20, \omega_c = 5)$.

The proof of proposition 2 is provided in Appendix D. According to (17), the CADRC exhibits reliable disturbance estimation and rejection capabilities for low-frequency disturbances. Moreover, as revealed by (18), in the same low-frequency limit, the transfer functions $L_{1,a}(jw)$ and $L_{2,a}(jw)$ approach zero more rapidly than those of LADRC, confirming the enhanced disturbance estimation and rejection capabilities.

To more intuitively illustrate the results of Proposition 2, the magnitude response curves of $L_{1,p}(s)$ and $L_{2,p}(s)$ are presented under multiple parameter configurations. The parameters of both CADRC and LADRC are configured as follows: the system order n ranges from 1 to 6, and the controller gains $k_j \triangleq C_n^{n-j+1}\omega_c^{n-j+1}(1 \leqslant j \leqslant n)$. Two distinct bandwidth scenarios are subsequently evaluated: (1) $\mu = 10, \omega_c = 10$; (2) $\mu = 20$, $\omega_c = 5$. As depicted in Figs. 2–5, the magnitude response curves of CADRC are uniformly lower than the LADRC ones at the same frequency, quantitatively demonstrating its superior capability in disturbance estimation and rejection.

V. Application to the Quanser 2-DOF Helicopter System

In this section, the experiments of a Quanser 2-DOF helicopter system are presented.

A diagram of the experimental setup is provided in Fig. 6. The Quanser 2-DOF helicopter system comprises a helicopter body equipped with a hardware-in-the-loop (HIL) control board, a yoke, two dc motors (pitch and yaw), two corresponding propellers, two encoders, and a slip ring. The pitch and yaw encoders measure the actual pitch and yaw angles of

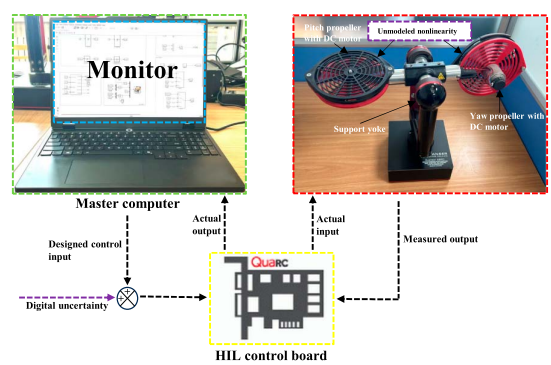

Fig. 6. Experimental setup of the quanser 2-DOF helicopter system.

the helicopter, respectively. Based on the implemented control algorithm, the master computer calculates the control input and sends it to the HIL control board. Subsequently, the HIL control board drives the pitch and yaw dc motors to regulate the corresponding angles. Additionally, the input voltages of both motors are limited to $[-24\,\mathrm{V}, 24\,\mathrm{V}]$ to protect the devices.

The mathematical model of the Quanser 2-DOF helicopter system is formulated as follows [34]:

$$\begin{cases} \left(I_p + ml_a^2\right)\ddot{\theta} = -mgl_a\cos\theta - D_p\dot{\theta} - ml_a^2\dot{\phi}^2\sin\theta\cos\theta \\ +K_{pp}\left(V_p + \triangle V_p\right) + K_{py}\left(V_y + \triangle V_y\right), \\ \left(I_y + ml_a^2\cos^2\theta\right)\ddot{\phi} = -D_y\dot{\phi} + 2ml_a^2\dot{\phi}\dot{\theta}\sin\theta\cos\theta \\ +K_{yp}\left(V_p + \triangle V_p\right) + K_{yy}\left(V_y + \triangle V_y\right), \end{cases} \tag{19}$$

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TABLE I
MAIN PARAMETERS OF THE 2-DOF HELICOPTER SYSTEM

Parameter	Value	Parameter	Value
l_a	$0.0071\,\mathrm{m}$	K_{pp}	$0.0012\,\mathrm{Nm/V}$
I_p	$0.0230\,\mathrm{kg}\cdot\mathrm{m}^2$	K_{py}	$0.0022\,\mathrm{Nm/V}$
I_y	$0.0210\,\mathrm{kg}\cdot\mathrm{m}^2$	K_{yp}	$-0.0027\,\mathrm{Nm/V}$
D_p	$0.0071\,\mathrm{N}\cdot\mathrm{s/m}$	K_{yy}	$0.0024\,\mathrm{Nm/V}$
D_y	$0.0220\,\mathrm{N}\cdot\mathrm{s/m}$	m	$1.0760\,\mathrm{kg}$

where $\theta(t) \in \mathbb{R}$ and $\phi(t) \in \mathbb{R}$ are the pitch angle and the yaw angle, $V_p(t) \in \mathbb{R}$ and $V_y(t) \in \mathbb{R}$ represent the voltages for the pitch and yaw motors, respectively. $\triangle V_p(t), \triangle V_y(t) \in \mathbb{R}$ are the designed digital uncertainties. The parameters l_a is the distance from the helicopter's mass center to the origin of the fixed fuselage frame, I_p and I_y represent the pitch and yaw moments of inertia, D_p and D_y are the corresponding friction coefficients, K_{pp}, K_{py}, K_{yp}, and K_{yy} are the torque-thrust gains, m is gross mass of helicopter. Moreover, the nominal values of $\{l_a, I_p, I_y, D_p, D_y, K_{pp}, K_{py}, K_{yp}, K_{yy}, m\}$ are summarized in Table I.

Based on the measurements of the attitude angle vector $p(t) = [p_1(t), p_2(t)]^\top \triangleq [\theta(t), \phi(t)]^\top$, the control target is to design the motor voltages $V(t) \triangleq [V_p(t), V_y(t)]^\top$ such that $p(t)$ tracks the given reference signal $r(t) = [r_1(t), r_2(t)]^\top \triangleq [30\,\mathrm{deg}, 30\,\mathrm{deg}]^\top$.

Denote $b_{11} = (K_{pp}/I_p + ml_a^2)$, $b_{12} = (K_{py}/I_p + ml_a^2)$, $g_1(\theta, \dot\theta, \phi, \dot\phi, t) = (-mgl_a\cos\theta - D_p\dot\theta - ml_a^2\dot\phi^2\sin\theta\cos\theta/I_p + ml_a^2) + (K_{pp}\triangle V_p + K_{py}\triangle V_y/I_p + ml_a^2)$, $b_{21}(t) = (K_{yp}/I_y + ml_a^2\cos^2\theta)$, $b_{22}(t) = (K_{yy}/I_y + ml_a^2\cos^2\theta)$, $g_2(\theta, \dot\theta, \phi, \dot\phi, t) = (-D_y\dot\phi + 2ml_a^2\dot\phi\dot\theta\sin\theta\cos\theta/I_y + ml_a^2\cos^2\theta) + (K_{yp}\triangle V_p + K_{yy}\triangle V_y/I_y + ml_a^2\cos^2\theta)$. Then, the proposed CADRC (6), (7), and (11) is applied to the Quanser 2-DOF helicopter with the parameters $(k_1 = 10, k_2 = 20, \mu = 12)$.

For comparative evaluation, the following traditional ADRC (TADRC) is also considered in the experiments:

$$\begin{cases} \dot{\tilde{x}}_{i1}(t) = \tilde{x}_{i2}(t) + 3\tilde{\mu}(p_i(t) - \tilde{x}_{i1}(t)), \\ \dot{\tilde{x}}_{i2}(t) = \tilde{x}_{i3}(t) + 3\tilde{\mu}^2(p_i(t) - \tilde{x}_{i1}(t)), \\ \dot{\tilde{x}}_{i3}(t) = \tilde{\mu}^3(p_i(t) - \tilde{x}_{i1}(t)), \quad i = 1, 2, \\ V_i^*(t) = \ddot{r}_i(t) - \tilde{k}_1(\tilde{x}_{i1}(t) - r_i(t)) \\ -\tilde{k}_2(\tilde{x}_{i2}(t) - \dot{r}_i(t)) - \tilde{x}_{i3}(t), \\ \tilde{V}(t) = B_v^{-1}(t)[V_1^*(t), V_2^*(t)]^\top \end{cases} \tag{20}$$

where $B_v(t) = (b_{ij}) \in \mathbb{R}^{2\times 2}$, and the controller parameters are set to $(\tilde{k}_1 = 10, \tilde{k}_2 = 20, \tilde{\mu} = 12)$ for a fair comparison.

To further demonstrate the superiority of the proposed control approach, we impose additional digital uncertainties $\triangle V_p(t)$ and $\triangle V_y(t)$ in Experiments 1 and 2.

Experiment 1: In this experiment, the additional digital uncertainties $\triangle V_p(t)$ and $\triangle V_y(t)$ are designed as sinusoidal signals combined with state-dependent nonlinearities

$$\text{Case 1}: \triangle V_p(t) = \triangle V_y(t) = 2\sin(2t) + 0.6\theta^3 + 0.4\phi^2,$$
$$\text{Case 2}: \triangle V_p(t) = \triangle V_y(t) = 4\sin(2t) + 0.6\theta^3 + 0.4\phi^2,$$

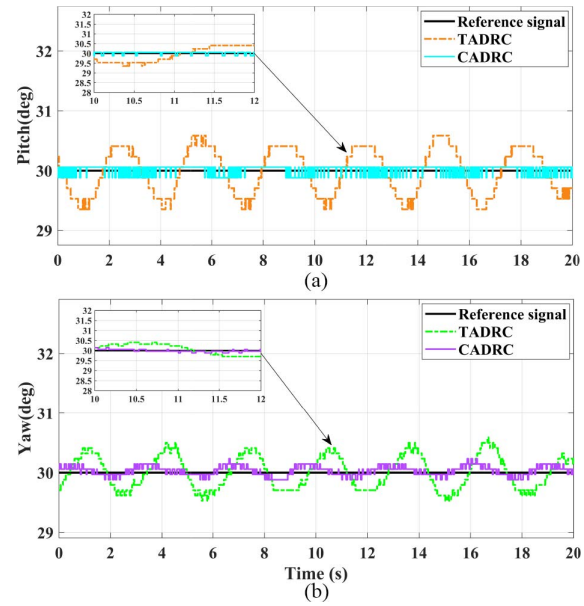

Fig. 7. Control performance under additional digital uncertainty (case 1). (a) Pitch. (b) Yaw.

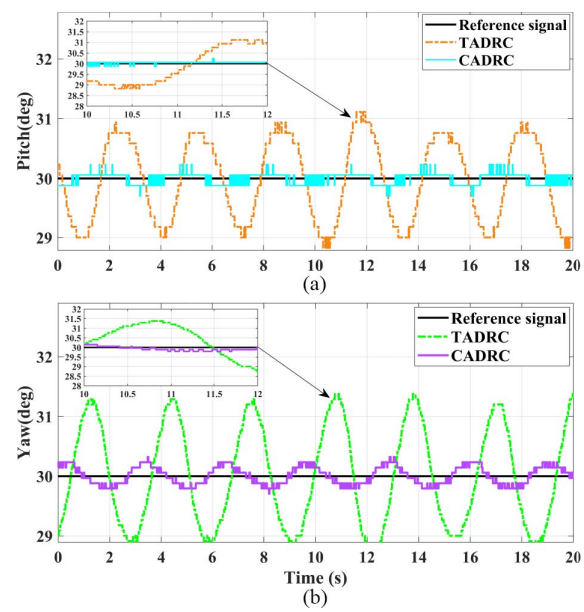

Fig. 8. Control performance under additional digital uncertainty (case 2). (a) Pitch. (b) Yaw.

$$\text{Case 3}: \triangle V_p(t) = \triangle V_y(t) = 8\sin(2t) + 0.6\theta^3 + 0.4\phi^2,$$
$$\text{Case 4}: \triangle V_p(t) = \triangle V_y(t) = 8\sin(4t) + 0.6\theta^3 + 0.4\phi^2.$$

Experiment 2: In this experiment, the additional digital uncertainties $\triangle V_p(t)$ and $\triangle V_y(t)$ are designed as step signals combined with state-dependent nonlinearities

$$\text{Case 5}: \triangle V_p(t) = \triangle V_y(t) = \begin{cases} 5 + \theta^2 + \phi, & \text{if } \sin(\pi t) \geqslant 0 \\ -5 + 0.5\phi^2, & \text{if } \sin(\pi t) < 0 \end{cases},$$
$$\text{Case 6}: \triangle V_p(t) = \triangle V_y(t) = \begin{cases} 7.5 + \theta^2 + \phi, & \text{if } \sin(\pi t) \geqslant 0 \\ -7.5 + 0.5\phi^2, & \text{if } \sin(\pi t) < 0 \end{cases},$$

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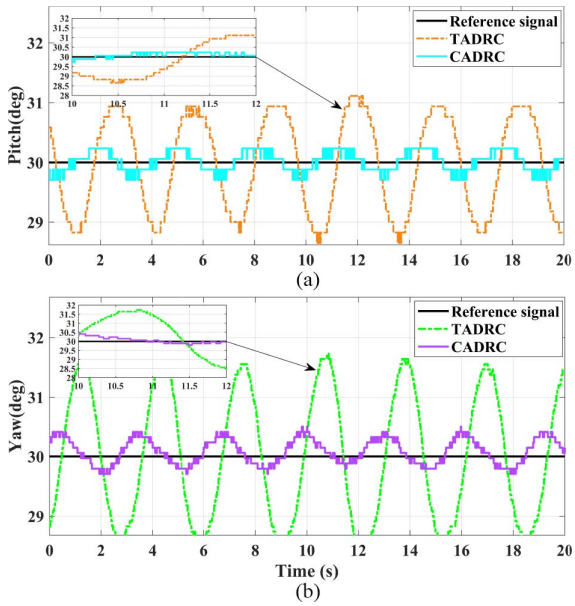

Fig. 9. Control performance under additional digital uncertainty (case 3). (a) Pitch. (b) Yaw.

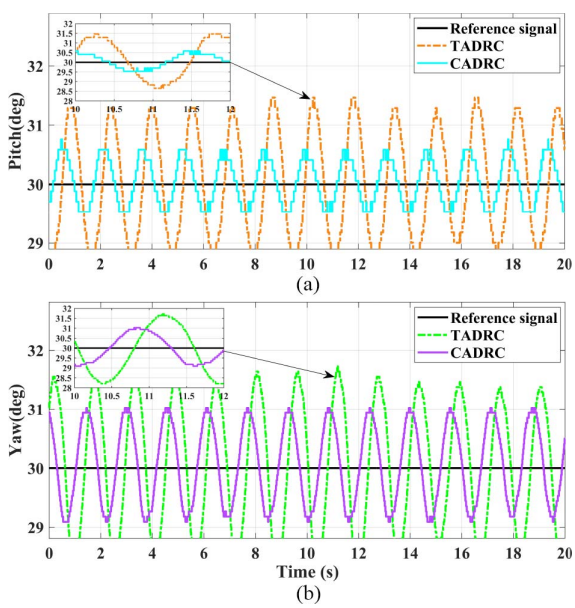

Fig. 10. Control performance under additional digital uncertainty (case 4). (a) Pitch. (b) Yaw.

$$\text{Case 7}: \triangle V_p(t) = \triangle V_y(t) = \begin{cases} 10 + \theta^2 + \phi, & \text{if } \sin(2\pi t) \geqslant 0 \\ -10 + 0.5\phi^2, & \text{if } \sin(2\pi t) < 0. \end{cases}$$

During the experiments, a sampling interval of 0.001 s was adopted, and the continuous-time controller was discretized via the forward Euler method. The results of the experiments are illustrated in Figs. 7–13. In addition, the integral of squared error (ISE), integral of absolute error (IAE), root mean square error (RMSE), and integral of squared control input (ISCI) are provided as key quantitative performance indicators in Tables II–III.

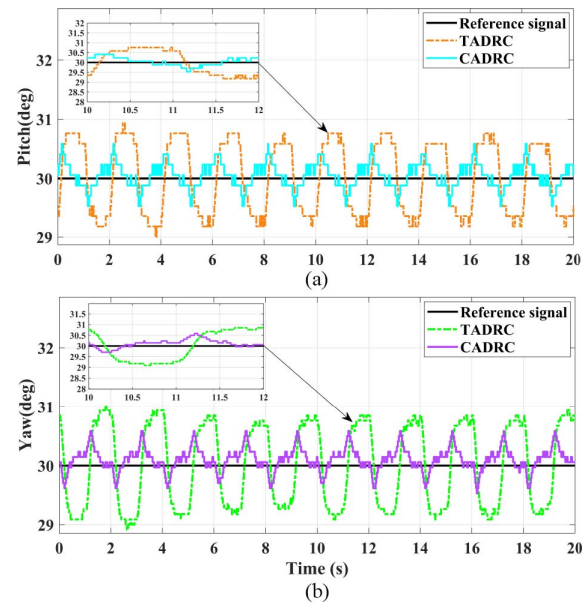

Fig. 11. Control performance under additional digital uncertainty (case 5). (a) Pitch. (b) Yaw.

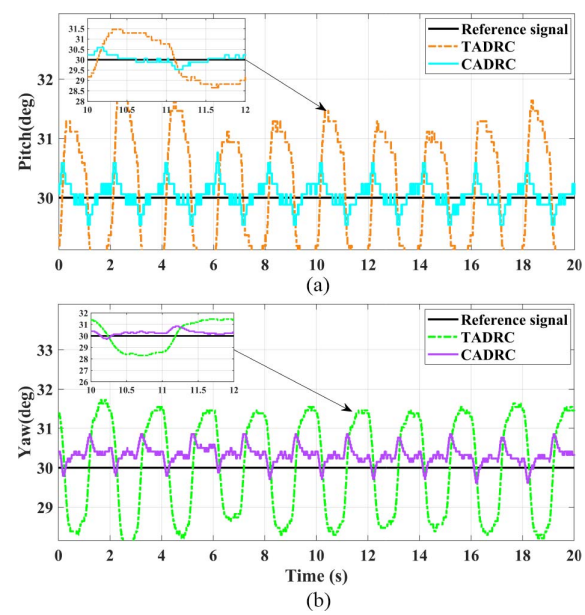

Fig. 12. Control performance under additional digital uncertainty (case 6). (a) Pitch. (b) Yaw.

As depicted in Figs. 7–13, the proposed CADRC delivers consistently superior pitch and yaw tracking performance over the traditional ADRC under all disturbance scenarios (Cases 1–7). For pitch angle control, CADRC achieves markedly lower ISE, IAE, and RMSE values compared to TADRC, with ISE reductions ranging from 82.48% (Case 4) to 98.14% (Case 2), IAE reductions from 58.52% (Case 4) to 86.31% (Case 2),

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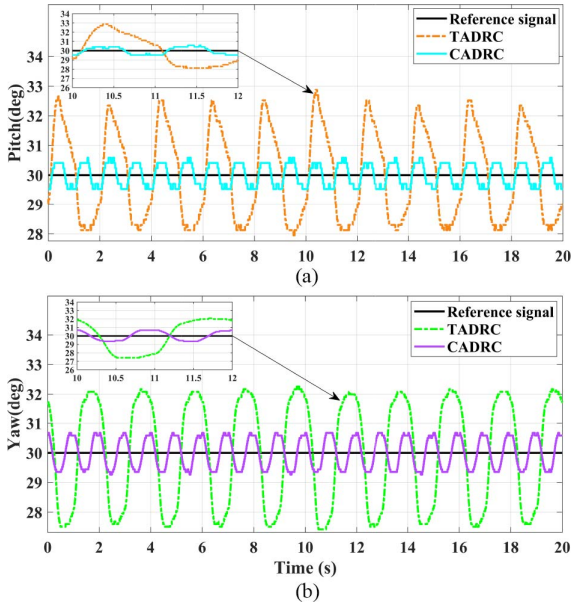

Fig. 13. Control performance under additional digital uncertainty (case 7). (a) Pitch. (b) Yaw.

TABLE II
PERFORMANCE INDICATORS OF PITCH ANGLE TRACKING FOR CASES 1–7

Test Type	Method	ISE	IAE	RMSE	ISCI
Case 1	CADRC	0.1247	1.4906	0.0790	18013
	TADRC	2.9863	6.9576	0.3864	17976
	RATIO	4.18%	21.42%	20.45%	100.21%
Case 2	CADRC	0.1786	1.7116	0.0945	17999
	TADRC	9.5771	12.5058	0.6920	17967
	RATIO	1.86%	13.69%	13.66%	100.18%
Case 3	CADRC	0.5807	2.9765	0.1704	18019
	TADRC	12.7338	14.5594	0.7979	17976
	RATIO	4.56%	20.44%	21.36%	100.24%
Case 4	CADRC	3.0425	7.0006	0.3900	18052
	TADRC	17.3676	16.8779	0.9319	18028
	RATIO	17.52%	41.48%	41.85%	100.13%
Case 5	CADRC	0.9602	3.6218	0.2191	18056
	TADRC	7.7053	11.7841	0.6207	18007
	RATIO	12.46%	30.73%	35.30%	100.27%
Case 6	CADRC	1.0614	3.6165	0.2304	18042
	TADRC	21.6500	19.5904	1.0404	18021
	RATIO	4.90%	18.46%	22.15%	100.12%
Case 7	CADRC	2.8517	7.0881	0.3776	18001
	TADRC	49.2088	28.9765	1.5686	18044
	RATIO	5.80%	24.46%	24.08%	99.76%

and RMSE reductions from 58.15% (Case 4) to 86.34% (Case 2). Similar advantages can be observed in yaw angle tracking control, where CADRC consistently outperforms TADRC, with ISE values reduced to only 3.19-36.10% of TADRC's values and RMSE reductions reaching up to 72.40% (Case 4). Critically, these significant accuracy improvements are achieved with nearly identical ISCI values (ISCI ratios: 99.65%–102.00%), demonstrating CADRC's efficient disturbance rejection capability.

TABLE III
PERFORMANCE INDICATORS OF YAW ANGLE TRACKING FOR CASES 1–7

Test Type	Method	ISE	IAE	RMSE	ISCI
Case 1	CADRC	0.1582	1.5011	0.0889	18013
	TADRC	1.5123	4.9016	0.2750	18009
	RATIO	10.46%	30.62%	32.33%	100.02%
Case 2	CADRC	0.4635	2.7180	0.1522	18013
	TADRC	14.5296	15.3194	0.8524	18028
	RATIO	3.19%	17.75%	17.85%	99.92%
Case 3	CADRC	1.0411	3.9038	0.2282	18103
	TADRC	23.8514	19.7283	1.0921	18046
	RATIO	4.36%	19.79%	20.90%	100.32%
Case 4	CADRC	9.1194	12.1800	0.6753	18084
	TADRC	25.2625	20.3145	1.1239	18034
	RATIO	36.10%	59.95%	60.09%	100.28%
Case 5	CADRC	1.0269	3.5973	0.2266	18126
	TADRC	10.0258	13.4549	0.7080	18006
	RATIO	10.24%	26.74%	32.01%	100.67%
Case 6	CADRC	3.0505	6.9116	0.3905	18388
	TADRC	33.5781	24.4416	1.2957	18028
	RATIO	9.09%	28.28%	30.14%	102.00%
Case 7	CADRC	5.2328	9.4609	0.5115	18006
	TADRC	68.6547	34.7738	1.8528	18069
	RATIO	7.62%	27.21%	27.60%	99.65%

VI. CONCLUSION

This article proposed a novel CFO-based ADRC for MIMO nonlinear uncertain systems. By leveraging an inductive decoupling-based Lyapunov method, the closed-loop stability was rigorously established under a broad class of state-dependent uncertainties. Moreover, a comparative frequency-domain analysis quantitatively demonstrated the superior disturbance estimation and rejection performance of the proposed method. Finally, experiments on a Quanser 2-DOF helicopter system confirmed that the proposed method outperforms the conventional ADRC.

In future work, we will focus on extending the CADRC framework to heterogeneous MIMO systems and stochastic SISO systems. Additionally, the improvement of the CFO structure to enhance robustness against measurement noise will be investigated to further facilitate practical applications.

APPENDIX

A. Proof of Lemma 1

To begin with, we define the estimation errors of (8) and (13) as $e_i(t) = x_{ie}(t) - \hat{x}_{ie}(t) \in \mathbb{R}^{n+1}$ and $\tilde{e}_i(t) = G_i(t) - \hat{G}_i(t) \in \mathbb{R}^{n+1}$, respectively. Denote their Laplace transforms as $E_i(s) = [E_{i,1}(s), \ldots, E_{i,n+1}(s)]^\top$ and $\tilde{E}_i(s) = [\tilde{E}_{i,1}(s), \ldots, \tilde{E}_{i,n+1}(s)]^\top$. In addition, let $X_{i(n+1)}(s)$, $\hat{X}_{i(n+1)}(s)$, and $\tilde{X}_{i(n+1)}(s)$ denote the Laplace transforms of the total disturbance $g_i(x(t), t)$, its estimations $\hat{x}_{i(n+1)}(t)$, and $\hat{g}_{i1}(t)$, respectively.

Due to (5), (8), and (13), $e_i(t)$ and $\tilde{e}_i(t)$ satisfy that $\dot{e}_i(t) = A_L e_i(t) + D_e(\dot{g}_i(x(t), t))$, $\dot{\tilde{e}}_i(t) = A_{\tilde{L}} \tilde{e}_i(t) + D_e(g_i^{(n+1)}(x(t), t))$, where $D_e = [0_{1\times n}, 1]^\top$, $A_L = A_e - D_e L_e^\top$, and $A_{\tilde{L}} = A_e - \tilde{L}_e C_e^\top$.

By taking the Laplace transform of $e_i(t)$, it can be obtained that $E_i(s) = (sI_{n+1} - A_L)^{-1} D_e(sX_{i(n+1)}(s))$,

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where $(sI_{n+1}-A_L)^{-1}D_e=[1,s,\ldots,s^{k-1},\ldots,s^n]\varrho$, and $\varrho=(1/s^{n+1}+l_1s^n+\cdots+l_ns+l_{n+1})$. Then, we can further obtain that $E_{i,n+1}(s)=(s^{n+1}/s^{n+1}+l_1s^n+\cdots+l_ns+l_{n+1})X_{i(n+1)}(s)$. By leveraging the equation $E_{i,n+1}(s)=X_{i(n+1)}(s)-\hat{X}_{i(n+1)}(s)$, it can be deduced that

$$\hat{X}_{i(n+1)}(s)=\frac{l_1s^n+\cdots+l_ns+l_{n+1}}{s^{n+1}+l_1s^n+\cdots+l_ns+l_{n+1}}X_{i(n+1)}(s). \tag{21}$$

Similarly, by taking the Laplace transform of $\tilde{e}_i(t)$, we obtain that $\tilde{E}_i(s)=(sI_{n+1}-A_{\tilde{L}})^{-1}D_e\left(s^{n+1}X_{i(n+1)}(s)\right)$, where $(sI_{n+1}-A_{\tilde{L}})^{-1}D_e=[1,s+l_1,\ldots,s^{k-1}+l_1s^{k-2}+\cdots+l_{k-1},\ldots,s^n+l_1s^{n-1}+\cdots+l_n]\varrho$, and $\varrho=(1/s^{n+1}+l_1s^n+\cdots+l_ns+l_{n+1})$. Then, we can further deduce that

$$\tilde{X}_{i(n+1)}(s)=\frac{l_1s^n+\cdots+l_ns+l_{n+1}}{s^{n+1}+l_1s^n+\cdots+l_ns+l_{n+1}}X_{i(n+1)}(s). \tag{22}$$

By Comparing (21) with (22), Lemma 1 is proved. □

B. Proof of Theorem 1

The proof of Theorem 1 proceeds in the following five steps.

1) Step1: The Establishment of Closed-Loop Error System: Let the tracking and estimation errors be defined as $E_i(t)=x_i(t)-x_i^*(t)\in\mathbb{R}^n$, $E(t)=[E_1^\top(t),\ldots,E_m^\top(t)]^\top\in\mathbb{R}^{nm}$, and $\xi_i(t)=T^{-1}\left(G_i(t)-\hat{G}_i(t)\right)\in\mathbb{R}^{n+1}$, $\xi(t)=[\xi_1^\top(t),\ldots,\xi_m^\top(t)]^\top\in\mathbb{R}^{(n+1)m}$, respectively, where the matrix $T\triangleq\mathrm{diag}\left\{\mu^{-n},\ldots,\mu^{-1},1\right\}\in\mathbb{R}^{(n+1)\times(n+1)}$.

Based on (5), (12), and (13), the dynamics of $(E_i(t),\xi_i(t))$ for $i\in\underline{m}$ are derived as follows:

$$\begin{cases}\dot{E}_i(t)=A_KE_i(t)+BC_e^\top T\xi_i(t),\\ \dot{\xi}_i(t)=\mu A_\phi\xi_i(t)+D_e\Psi_i(E(t),\xi(t),t),\end{cases} \tag{23}$$

where $C_e=[1,0_{1\times n}]^\top$, $D_e=[0_{1\times n},1]^\top$, $\Psi_i(E(t),\xi(t),t)=(d^{n+1}f_i/dt^{n+1})$, and the matrices $A_K\in\mathbb{R}^{n\times n}$, $A_\phi\in\mathbb{R}^{(n+1)\times(n+1)}$ have the following form:

$$A_K=\begin{bmatrix}0&1&\cdots&0\\ \vdots&\vdots&\ddots&\vdots\\ 0&0&\cdots&1\\ -k_1&-k_2&\cdots&-k_n\end{bmatrix},\ A_\phi=\begin{bmatrix}-C_{n+1}^1&1&\cdots&0\\ \vdots&\vdots&\ddots&\vdots\\ -C_{n+1}^n&0&\cdots&1\\ -C_{n+1}^{n+1}&0&\cdots&0\end{bmatrix}.$$

2) Step2: The Analysis of the Coupling Effect Caused by the Total Disturbances: To begin with, it can be noticed that $(d^{n+1}/dt^{n+1})g_i(x(t),t)=(d^{n+1}/dt^{n+1})\delta_i(x(t),t)+(d^{n+1}/dt^{n+1})d_i(t)$, and $C_e^\top T=\mu^{-n}C_e$, $C_e^\top TA_\phi^jD_e=0$ for $0\leqslant j\leqslant n-1$. Subsequently, it will be proved that, for any positive integer $1\leqslant k\leqslant n+1$, there exist functions $\vartheta_{l_k}(\cdot)$ and $\vartheta_{l_{k,z}}(\cdot)\in\mathcal{C}^1(\mathbb{R}^n\times\mathbb{R},\mathbb{R})$ such that

$$\frac{\mathrm{d}^k\delta_i}{\mathrm{d}t^k}=\vartheta_{l_k}(x,t)+\sum_{z=0}^{k-1}\sum_{L_{k,z}\in\varpi(k,z)}\vartheta_{L_{k,z}}(x,t)\left[\prod_{p=1}^m\prod_{q=0}^z\left(\frac{C_e^\top A_\phi^q\xi_p(t)}{\mu^n}\right)^{l_{p,q}^{(k)}}\right]\mu^z \tag{24}$$

where $L_{k,z}=(l_{1,0}^{(k)},\ldots,l_{1,z}^{(k)},\ldots,l_{m,0}^{(k)},\ldots,l_{m,z}^{(k)})\in\mathbb{N}^{mz+m}$ denotes the multi-index notation, and the index set $\varpi(k,z)$ is defined as $\varpi(k,z)\triangleq\{(l_{1,0},\ldots,l_{1,z},\ldots,l_{m,0},\ldots,l_{m,z})\in\mathbb{N}^{mz+m}|\ 0\leqslant l_{p,q}\leqslant k,\ 1\leqslant p\leqslant m,\ 0\leqslant q\leqslant z,\ \sum_{q=1}^z(q\sum_{p=1}^m l_{p,q})=z\}$.

The proof proceeds by induction on the derivative order k. For the case $k=1$, it can be deduced that $(\mathrm{d}\delta_i/\mathrm{d}t)=\vartheta_{l_1}(x(t),t)+\sum_{p=1}^m\vartheta_{l_{p,0}}(x(t),t)\mu^{-n}C_e^\top\xi_p(t)$, where $\vartheta_{l_{p,0}}(x,t)=(\partial\delta_i/\partial x_{pn})$, $p=1,2,\ldots,m$, and $\vartheta_{l_1}(x(t),t)=\sum_{j=1}^m(\partial\delta_i/\partial x_j)\left(A_Kx_j(t)+Br_j^{(n)}(t)+BK^\top\bar{r}_j(t)\right)+(\partial\delta_i/\partial t)$.

Assuming that (24) holds for any positive integer $k=s$ $(\forall 1\leqslant s\leqslant n)$, it is obtained that $(\mathrm{d}(\mu^{-n}C_e^\top A_\phi^q\xi_p(t))^{l_{p,q}}/\mathrm{d}t)=(l_{p,q}\mu)(\mu^{-n}C_e^\top A_\phi^q\xi(t))^{l_{p,q}-1}(\mu^{-n}C_e^\top A_\phi^{q+1}\xi_p(t))$, $(\mathrm{d}\vartheta_{L_{k,z}}(x,t)/\mathrm{d}t)=\sum_{j=1}^m(\partial\vartheta_{L_{k,z}}/\partial x_j)(A_Kx_j(t)+Br_j^{(n)}(t)+BK^\top\bar{r}_j(t))+(\partial\vartheta_{L_{k,z}}/\partial t)+\sum_{j=1}^m(\partial\vartheta_{L_{k,z}}/\partial x_{jn})C_e^\top T\xi_j(t)$. Then, we further verify that (24) holds for $k=s+1$.

Given the fact that the functions $\vartheta_{l_k}(x,t)$, $\vartheta_{L_{k,z}}(x,t)$ in (24) are composed of finite products and sums of $(x_p,x_p^*,r_p^{(i)},(\partial^{i_x+i_t}\delta_i/\partial x^{i_x}\partial t^{i_t}))$ with $1\leqslant p\leqslant m$, $1\leqslant i\leqslant 2n+1$, and $1\leqslant i_x+i_t\leqslant n+1$, along with Assumptions 1–3, we can obtain that there exists a non-decreasing function $\Phi(\cdot)$ so that $\sup_{L_{k,z}\in\varpi(k,z)}\{|\vartheta_{l_k}(x(t),t)|,|\vartheta_{L_{k,z}}(x(t),t)|\}\leqslant\Phi(\rho_e)$, for any $E(t)\in\{E\in\mathbb{R}^{nm}:||E||\leqslant\rho_e\}$.

Finally, it is deduced that if $||E(t)||\leqslant\rho_e$, $||\xi(t)||\leqslant\mu^r\rho_\xi$ $(\forall 0\leq r\leq n)$, then

$$|\Psi_i(E(t),\xi(t),t)|\leqslant\sum_{j=0}^r N_{i,r,j}\mu^j \tag{25}$$

where $N_{i,r,0}=N_d+\Phi(\rho_e)+\sum_{z=0}^{n-r}\sum_{L_{n+1,z}\in\varpi(n+1,z)}\Phi(\rho_e)[\prod_{p=1}^m\prod_{q=0}^z(||A_\phi||^q\rho_\xi)^{l_{p,q}}]$, $N_{i,r,j}=\sum_{L_{n+1,z}\in\varpi(n+1,z)}\Phi(\rho_e)[\prod_{p=1}^m\prod_{q=0}^z(||A_\phi||^q\rho_\xi)^{l_{p,q}}]$ $(1\leqslant j\leqslant r,\ z=n+j-r)$.

3) Step3: The Boundedness Analysis of the Trajectories of the Closed-Loop Error system: Since the matrices A_K and A_ϕ are Hurwitz, there exist unique positive definite matrices Q_K and Q_ϕ such that $A_K^\top Q_K+Q_KA_K=-I_n$ and $A_\phi^\top Q_\phi+Q_\phi A_\phi=-I_{n+1}$. Define the Lyapunov functions $V_{i1}(t)=E_i^\top(t)Q_KE_i(t)$ and $V_{i2}(t)=\xi_i^\top(t)Q_\phi\xi_i(t)$, $i\in\underline{m}$. Moreover, it can be verified that

$$\begin{cases}\lambda_{k1}||E_i(t)||^2\leqslant V_{i1}(t)\leqslant\lambda_{k2}||E_i(t)||^2,\\ \lambda_{\phi1}||\xi_i(t)||^2\leqslant V_{i2}(t)\leqslant\lambda_{\phi2}||\xi_i(t)||^2\end{cases} \tag{26}$$

where λ_{k1} and λ_{k2} represent the minimal and maximal eigenvalues of Q_K, $\lambda_{\phi1}$ and $\lambda_{\phi2}$ represent the minimal and maximal eigenvalues of Q_ϕ, respectively.

Then, it will be proved that there exists a positive constant $\tilde{\mu}>1$ such that $(E_1(t),\ldots,E_m(t),\xi_1(t),\ldots,\xi_m(t))$ remains in the invariant set: $\Omega\triangleq\{(E_1,\ldots,E_m,\xi_1,\ldots,\xi_m)\in\mathbb{R}^{(2n+1)m}:V_{i1}\leqslant\rho_{k1},V_{i2}\leqslant\rho_{\phi1},i\in\underline{m}\}$, for any $\mu\geqslant\tilde{\mu}$, where ρ_{k1}, $\rho_{\phi1}$ is defined by $\rho_0\triangleq 2\max_{1\leqslant i\leqslant m}\{F_i(t_0)-\hat{F}_i(t_0)\}>0$, $\rho_{k1}\triangleq(9\lambda_{k2}\lambda_{\phi2}/\lambda_{\phi1})\|Q_K\|^2\rho_0^2$, $\rho_{\phi1}\triangleq 2\lambda_{\phi2}\rho_0^2\mu^{2n}$.

The proof consists of the following two steps.

(S1) For any integer $1\leqslant i\leqslant m$, suppose that there exists a positive constant $t_i^*\in(t_0,+\infty)$ such that $V_{i2}(t_i^*)=\rho_{\phi1}$, $\max_{1\leqslant j\leqslant m,\ j\neq i}V_{j2}(t)\leqslant\rho_{\phi1}$, $\max_{1\leqslant j\leqslant m}V_{j1}(t)\leqslant\rho_{k1}$, $t\in[t_0,t_i^*]$, which implies that $(\rho_{\phi1}/\lambda_{\phi2})\leqslant||\xi_i(t_i^*)||^2\leqslant$

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$(\rho_{\phi1}/\lambda_{\phi1})$, and $\|\xi(t_i^*)\| \leqslant \mu^n\sqrt{(2m\lambda_{\phi2}\rho_0^2/\lambda_{\phi1})}$, $\|E(t_i^*)\| \leqslant \sqrt{(m\rho_{k1}/\lambda_{k1})}$.

Then, according to (23) and (25), the derivative of $V_{i2}(t_1^*)$ satisfies that $(\mathrm{d}V_{i2}(t_1^*)/\mathrm{d}t) = -\mu\|\xi_i(t_1^*)\|^2 + 2(D_e^\top Q_\phi\xi_i(t_1^*))\Psi_i(E(t_1^*), \xi(t_1^*), t_1^*) \leqslant \|\xi_1(t_1^*)\|(-\sqrt{2}\rho_0\mu^{n+1} + 2\|Q_\phi\|(\sum_{j=0}^n N_{i,n,j}\mu^j))$, where $N_{i,n,j}(0 \leqslant j \leqslant n)$ are positive constants independent of μ. Based on polynomial theory, there exists a positive constant $\tilde{\mu}_i > 1$ such that $(\mathrm{d}V_{i2}(t_1^*)/\mathrm{d}t) < 0$.

(S2) For any integer $1 \leqslant i \leqslant m$, suppose that there exists a positive constant $\tilde{t}_i \in (t_0, +\infty)$ such that $V_{i1}(\tilde{t}_i) = \rho_{k1}$, $\max_{1\leqslant j\leqslant m} V_{j2}(t) \leqslant \rho_{\phi1}$, $\max_{\substack{1\leqslant j\leqslant m,\\ j\neq i}} V_{j1}(t) \leqslant \rho_{k1}$, $t \in [t_0, \tilde{t}_i]$, which implies that $(\rho_{k1}/\lambda_{k2}) \leqslant \|E_i(\tilde{t}_i)\|^2 \leqslant (\rho_{k1}/\lambda_{k1})$, and $\|\xi_i(\tilde{t}_i)\|^2 \leqslant (\rho_{\phi1}/\lambda_{\phi1})$.

Then, due to (23), the derivative of $V_{i1}(\tilde{t}_1)$ satisfies that $(\mathrm{d}V_{i1}(\tilde{t}_1)/\mathrm{d}t) = -\|E_i(\tilde{t}_i)\|^2 + 2E_i^\top(\tilde{t}_i)Q_K BC_e^\top T\xi_i(\tilde{t}_i) \leqslant \|E_i(\tilde{t}_i)\|(2\sqrt{2}-3)\|Q_K\|\rho_0\sqrt{\lambda_{\phi2}}/\sqrt{\lambda_{\phi1}}) < 0$.

By defining $\tilde{\mu} = \max_{1\leqslant j\leqslant m}\tilde{\mu}_j$, and combining (S1) and (S2), we can deduce that $(E_1(t), \ldots, E_m(t), \xi_1(t), \ldots, \xi_m(t))$ remains in the invariant set Ω for any $\mu \geqslant \tilde{\mu}$ and $t \geqslant t_0$. Moreover, we define $\rho_{1,e} = \sqrt{\rho_{k1}/\lambda_{k1}}$ and $\rho_{1,\xi} = \sqrt{\rho_{\phi1}/\lambda_{\phi1}}$.

4) Step4: The Further Analysis of the Upper Bound for the Estimation Errors $\xi(t)$: By leveraging (23), we get that $(\mathrm{d}\sqrt{V_{i2}(t)}/\mathrm{d}t) \leqslant -(\mu/2\lambda_{\phi2})\sqrt{V_{i2}(t)} + (\|Q_\phi\|/\sqrt{\lambda_{\phi1}})|\Psi_i(E(t), \xi(t), t)|$. Then, based on Gronwall inequality and (25), the bound of $\sqrt{V_{i2}(t)}$ is given by $\sqrt{V_{i2}(t)} \leqslant (\sqrt{\lambda_{\phi2}}\rho_0)\mu^n e^{-(\mu/2\lambda_{\phi2})(t-t_0)} + (\|Q_\phi\|/\sqrt{\lambda_{\phi1}})N_{i,n,\max}\mu^n\int_{t_0}^t e^{-(\mu/2\lambda_{\phi2})(t-\tau)}\mathrm{d}\tau$, where $N_{i,n,\max} \triangleq \max_{0\leqslant j\leqslant n} N_{i,n,j}$, and $N_{i,n,j}$ $(0 \leqslant j \leqslant n)$ are positive constants independent of μ.

Let $t_1 = t_0 + 2\lambda_{\phi2}(ln\mu/\mu)$, and it can be derived that $\|\xi_i(t)\| \leqslant (\sqrt{V_{i2}(t)}/\sqrt{\lambda_{\phi1}}) \leqslant \mu^{n-1}\rho_{i1}^*$, $\forall t \geqslant t_1$, where $\rho_{i1}^* = ((\sqrt{2\lambda_{\phi2}}\rho_0/\sqrt{\lambda_{\phi1}}) + (2N_{i,n,\max}\|Q_\phi\|\lambda_{\phi2}/\lambda_{\phi1})$. Then, we can obtain that $\|\xi(t)\| = \sqrt{\sum_{i=1}^m\|\xi_i(t)\|^2} \leqslant \mu^{n-1}\rho_1^*$, $\forall t \geqslant t_1$, where $\rho_1^* = \sqrt{\sum_{i=1}^m(\rho_{i1}^*)^2}$. In conjunction with (25), we know that $|\Psi_i(E(t), \xi(t), t)| \leqslant \sum_{j=0}^{n-1} N_{i,n-1,j}\mu^j$, $\forall t \geqslant t_1$, where $N_{i.n-1,j}$ $(0 \leqslant j \leqslant n-1)$ are independent positive constants of μ.

Furthermore, according to (23) and Gronwall inequality, $\sqrt{V_{i2}(t)}$ has the following bound: $\sqrt{V_{i2}(t)} \leqslant e^{-(\mu/2\lambda_{\phi2})(t-t_1)}\sqrt{V_{i2}(t_1)} + \int_{t_1}^t e^{-(\mu/2\lambda_{\phi2})(t-\tau)}(\|Q_\phi\|/\sqrt{\lambda_{\phi1}})|\Psi_i(E(\tau), \xi(\tau), \tau)|\mathrm{d}\tau$, $\forall t \geqslant t_1$. Similar to before, by defining $t_2 = t_0 + 4\lambda_{\phi2}(ln\mu/\ \mu)$, it can be concluded that $\|\xi(t)\| \leqslant \mu^{n-2}\rho_{i2}^*$, $\forall t \geqslant t_2$, where $\rho_{i2}^* = ((\sqrt{\lambda_{\phi2}}\rho_1^*/\sqrt{\lambda_{\phi1}}) + (2N_{i,n-1,\max}\|Q_\phi\|\lambda_{\phi2}/\lambda_{\phi1}))$. Next, we can further obtain that $\|\xi(t)\| \leqslant \mu^{n-2}\rho_2^*$, $\forall t \geqslant t_2$, where $\rho_2^* = \sqrt{\sum_{i=1}^m(\rho_{i2}^*)^2}$.

In conclusion, by defining $\tilde{t} = t_0 + 2n\lambda_{\phi2}(ln\mu/\mu)$ and using mathematical induction, we can prove that there exists a positive constant ρ_n^* such that $\|\xi(t)\| \leqslant \rho_n^*$, $\forall t \geqslant \tilde{t}$, where ρ_n^* is independent of μ and $\lim_{\mu\to+\infty}\tilde{t} = t_0$.

5) Step5: The Convergence Analysis of the Trajectories of Closed-Loop Error System: By leveraging (25), the bound for $\Psi_i(E(t), \xi(t), t)$ is given by: $|\Psi_i(E(t), \xi(t), t)| \leqslant \pi_i(\rho_{1,e}, \rho_n^*)$, $\forall t \geqslant \tilde{t}$, for any $\mu \geqslant \tilde{\mu}$, where $\pi_i(\cdot)$ is a well-defined function independent of μ. Then, we define $\pi(\rho_{1,e}, \rho_n^*) = \max_{1\leqslant i\leqslant m}\pi_i(\rho_{1,e}, \rho_n^*)$.

By using the dynamics (23) and the Gronwall inequality, it can be derived that $\sqrt{V_{i2}(t)} \leqslant e^{-(\mu/2\lambda_{\phi2})(t-\tilde{t})}\sqrt{V_{i2}(\tilde{t})} + \int_{\tilde{t}}^t e^{-(\mu/2\lambda_{\phi2})(t-\tau)}(\|Q_\phi\|/\sqrt{\lambda_{\phi1}})|\Psi_i(E(\tau), \xi(\tau), \tau)|\mathrm{d}\tau$. Then, it can be obtained that $\|G_i(t) - \hat{G}_i(t)\| \leqslant \rho_2(e^{-\rho_3\mu(t-\tilde{t})} + (1/\mu))$, $\forall t \geqslant \tilde{t}$, where $\rho_2 = \max\{(\sqrt{\lambda_{\phi2}}\rho_n^*/\sqrt{\lambda_{\phi1}}), (2\|Q_\phi\|\lambda_{\phi2}\pi(\rho_{1,e}, \rho_n^*)/\sqrt{\lambda_{\phi1}})\}$, $\rho_3 = (1/2\lambda_{\phi2})$. Subsequently, we analyze the convergence of $\sqrt{V_{i1}(t)}$. For all $t \geqslant \tilde{t}$, we can deduce that $(\mathrm{d}\sqrt{V_{i1}(t)}/\mathrm{d}t) \leqslant -(1/2\lambda_{k2})\sqrt{V_{i1}(t)} + (\|Q_K\|/\sqrt{\lambda_{k1}})\|\xi_i(t)\|\mu^{-n}$, $\forall t \geqslant \tilde{t}$. By leveraging Gronwall inequality, the upper bound for $\sqrt{V_{i1}(t)}$ can be established as: $\sqrt{V_{i1}(t)} \leqslant e^{-(1/2\lambda_{k2})(t-\tilde{t})}\sqrt{\lambda_{k2}}\|E_i(\tilde{t})\| + (2\lambda_{k2}\|Q_K\|\rho_n^*/\sqrt{\lambda_{k1}})(1/\mu^n)$, $\forall t \geqslant \tilde{t}$.

Finally, from (23), it can be derived that $\|E_i(t)\| \leqslant \rho_0^*(ln\mu/\mu)$, $\forall t_0 \leqslant t \leqslant \tilde{t}$, where $\rho_0^* = 2n^2\lambda_{\phi2}((3\sqrt{\lambda_{k2}\lambda_{\phi2}}/\sqrt{\lambda_{k1}\lambda_{\phi1}})\|A_K\|\|Q_K\| + (\sqrt{2\lambda_{\phi2}}/\sqrt{\lambda_{\phi1}}))\rho_0$. Then, we further obtain that for any $\mu \geqslant \tilde{\mu}$, and $i \in \underline{m}$, $\|E_i(t)\| \leqslant \rho_1(ln\mu/\mu)$, $\forall t \geqslant t_0$, where $\rho_1 = \max\{\rho_0^*, \sqrt{\lambda_{k2}}\rho_0^* + (2\lambda_{k2}\|Q_K\|\rho_n^*/\sqrt{\lambda_{k1}})\}$. □

C. Proof of Proposition 1

Taking Laplace transforms of (5) and (11), we derive that $sX_i(s) = AX_i(s) + B(U_i^*(s) + X_{i(n+1)}(s))$, $U_i^*(s) = -K^\top X_i(s) + (s^n + \sum_{p=0}^{n-1}k_{n-p}s^{n-1-p})R(s) - \hat{X}_{i(n+1)}(s)$, and $Y_i(s) = C^\top X_i(s)$.

Then, it can be deduced that $Y_i(s) = C^\top(sI_n - A_K)^{-1}BD_e^\top(X_{i(n+1)}(s) - \hat{X}_{i(n+1)}(s)) + C^\top(sI_n - A_K)^{-1}B(s^n + \sum_{p=0}^{n-1}k_{n-p}s^{n-1-p})R(s)$, where $(sI_n - A_K)^{-1}B = [1, s, \ldots, s^{n-1}]^\top\iota$, and $\iota = (1/(s^n + \sum_{p=0}^{n-1}k_{n-p}s^{n-1-p}))$.

In conjunction with (21), we obtain that $Y_i(s) = R_i(s) + (s^{n+1}/(s^{n+1} + \sum_{p=1}^{n+1}l_p s^{n+1-p})(s^n + \sum_{p=0}^{n-1}k_{n-p}s^{n-1-p}))X_{i(n+1)}$. Furthermore, we denote $L_{1,a}(s) = (s^{n+1}/s^{n+1} + \sum_{p=1}^{n+1}l_p s^{n+1-p})$ and $L_{2,a}(s) = (s^{n+1}/(s^{n+1} + \sum_{p=1}^{n+1}l_p s^{n+1-p})(s^n + \sum_{p=0}^{n-1}k_{n-p}s^{n-1-p}))$.

Similar to the derivation of $L_{1,a}(s)$ and $L_{2,a}(s)$, the expressions of $L_{1,b}(s)$ and $L_{2,b}(s)$ corresponding to LADRC can be found in [33]. □

D. Proof of Proposition 2

From Proposition 1, the following equations are obtained:

$$\begin{cases} \lim\limits_{w\to0}|L_{1,a}(jw)| = 0, \quad \lim\limits_{w\to0}|L_{2,a}(jw)| = 0, \\ \frac{L_{1,a}(s)}{L_{1,b}(s)} = \frac{s^n}{s^n + \sum_{p=1}^n l_p s^{n-p}}, \\ \frac{L_{2,a}(s)}{L_{2,b}(s)} = \frac{s^n}{s^n + \sum_{p=1}^n(l_p + \sum_{q=1}^{p-1}k_{n+1-q}l_{p-q} + k_{n+1-p})s^{n-p}}. \end{cases} \tag{27}$$

Then, it follows that $\lim_{w\to0}|(L_{1,a}(jw)/L_{1,b}(jw))| = 0$ and $\lim_{w\to0}|(L_{2,a}(jw)/L_{2,b}(jw))| = 0$, thereby verifying Proposition 2. □

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